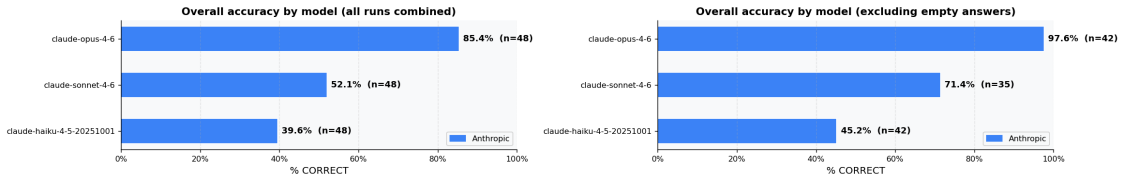
results_analysis_v2

March 13, 2026

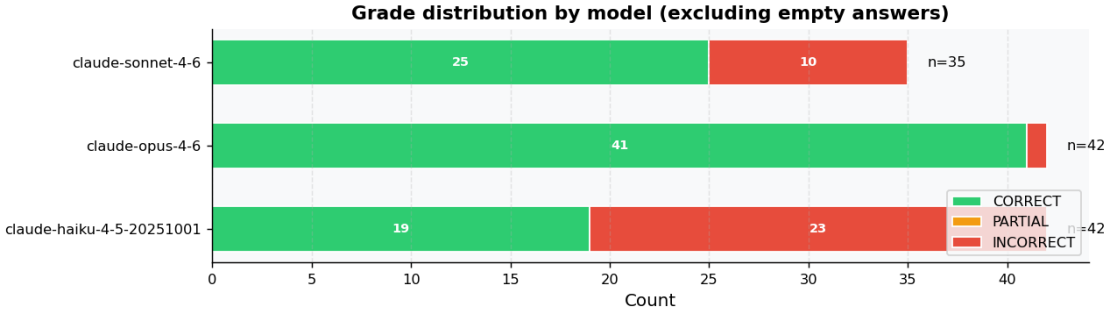
1 TinyBench — Evaluation Results Analysis

```
Loaded 4 result file(s):
  ../data/results/run_03_12_2026/by_template_python_only/anomaly_data_quality_fi
lter_v0.json  →  run='python_only'
  ../data/results/run_03_12_2026/by_template_python_only/anomaly_data_quality_fi
lter_v1.json  →  run='python_only'
  ../data/results/run_03_12_2026/by_template_python_semantic/anomaly_data_qualit
y_filter_v0.json  →  run='semantic'
  ../data/results/run_03_12_2026/by_template_python_semantic/anomaly_data_qualit
y_filter_v1.json  →  run='semantic'

144 total records  |  3 models  |  2 runs  |  2 templates
Empty answers: 25
Records after excluding empty answers: 119 / 144
Runs present: ['python_only', 'semantic']
```

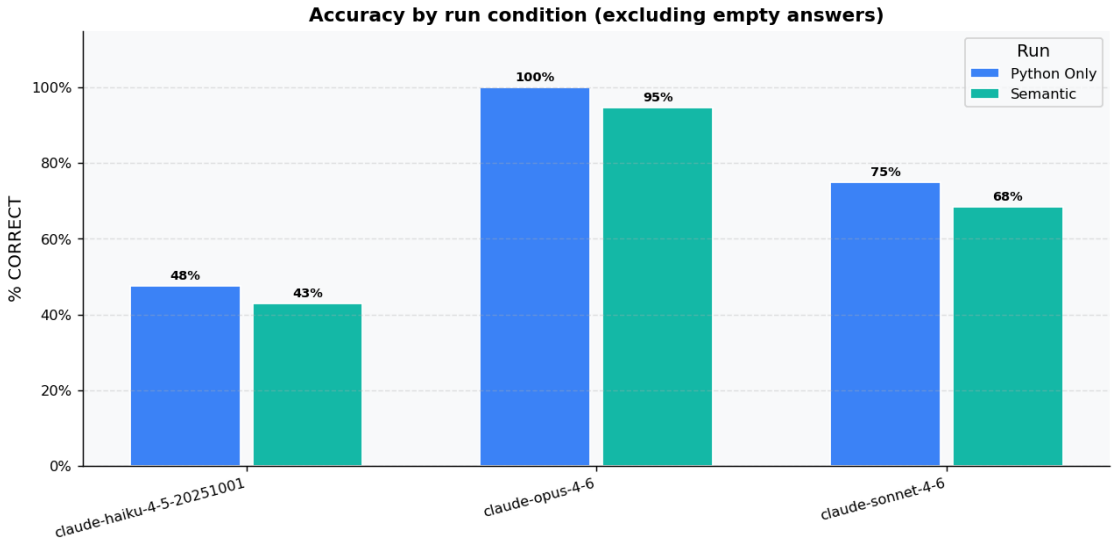
1.1 1 — Overall Accuracy by Model



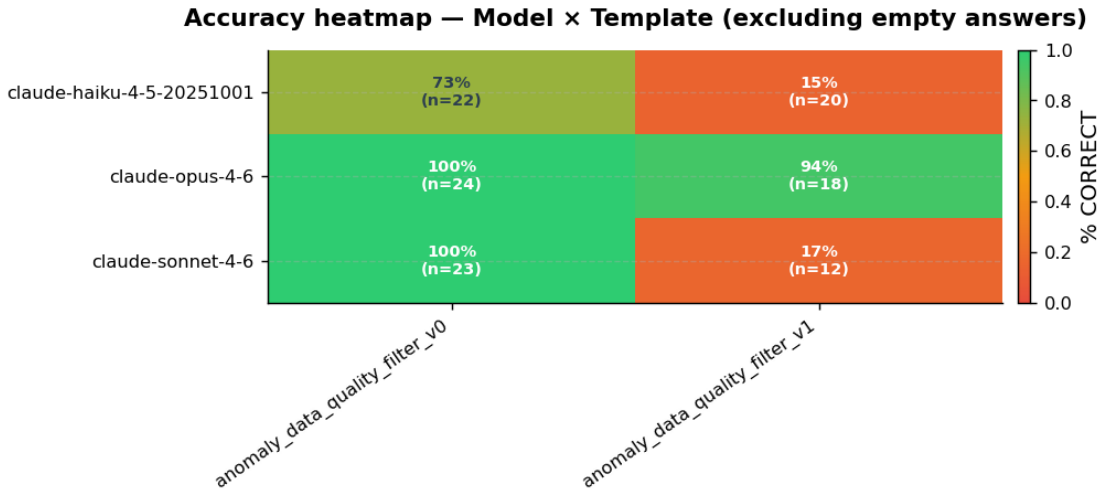
1.2 2 — Grade Distribution



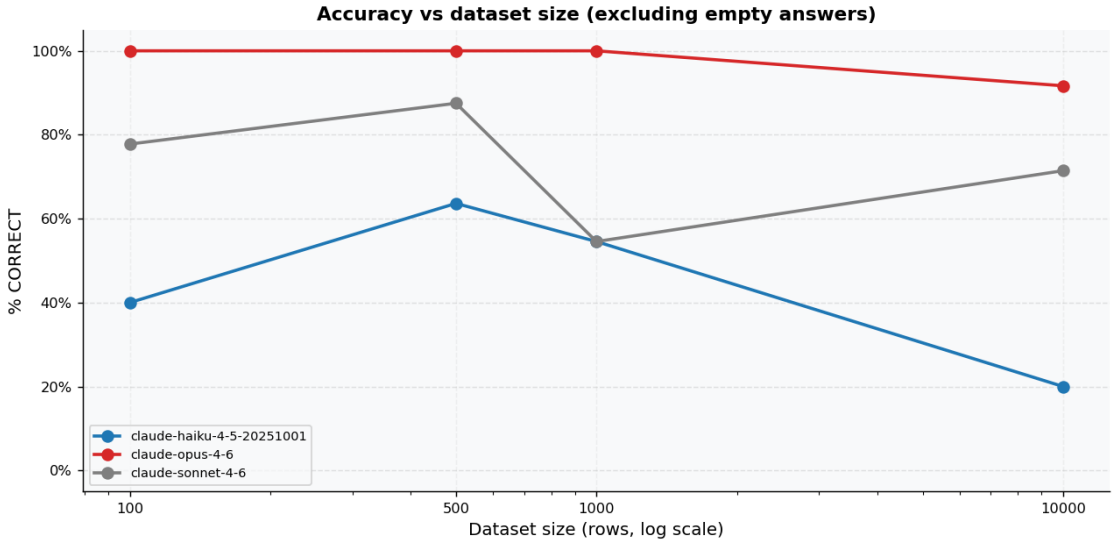
1.3 3 — Accuracy by Run Condition



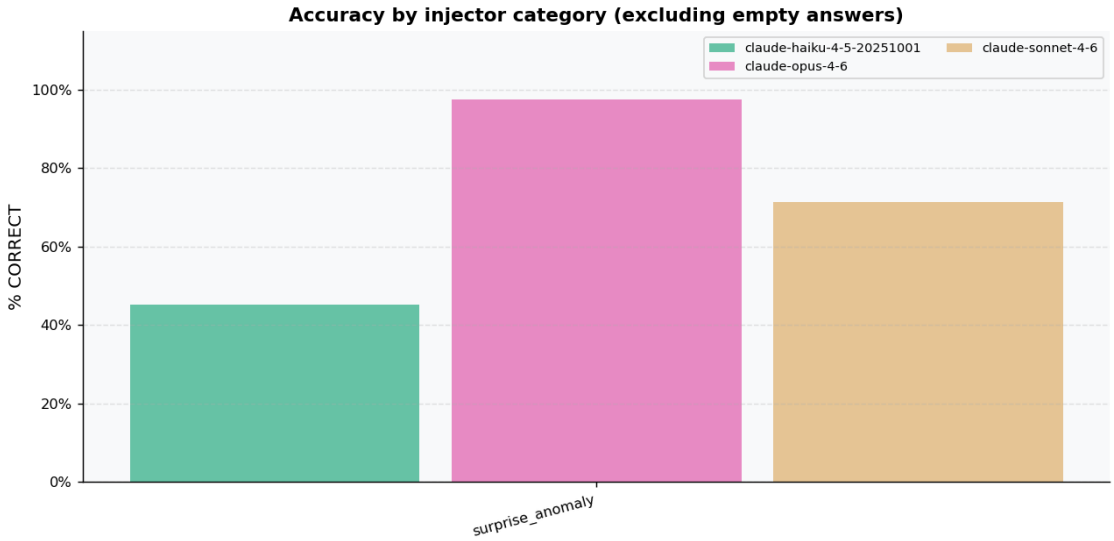
1.4 4 — Accuracy Heatmap: Model × Template



1.5 5 — Accuracy by Dataset Size

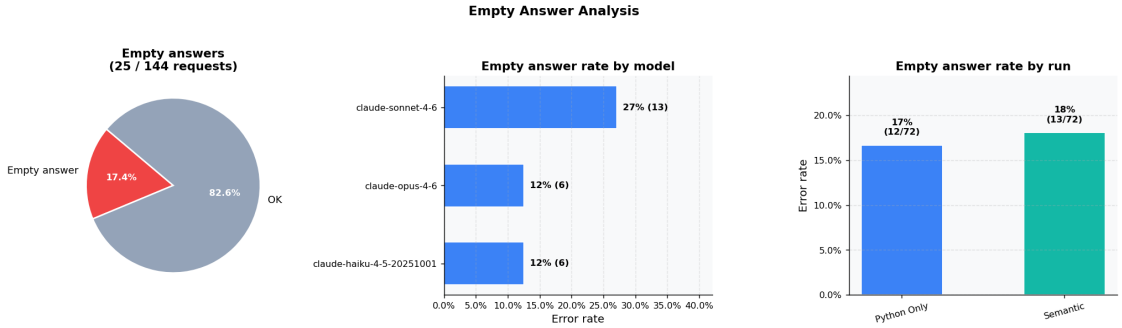


1.6 6 — Accuracy by Injector Category



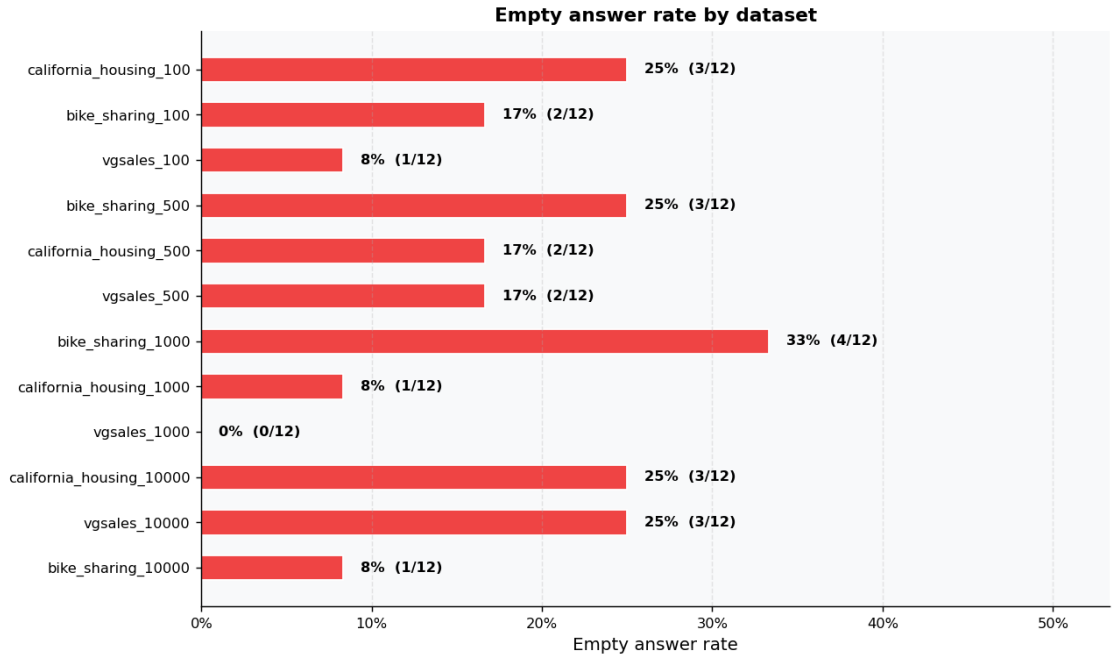
1.7 7 — Empty Answer Analysis

```
Empty answers: 25 / 144 (17.4%)
Affected datasets: ['bike_sharing_100', 'bike_sharing_1000',
'bike_sharing_10000', 'bike_sharing_500', 'california_housing_100',
'california_housing_1000', 'california_housing_10000', 'california_housing_500',
'vgsales_100', 'vgsales_10000', 'vgsales_500']
```

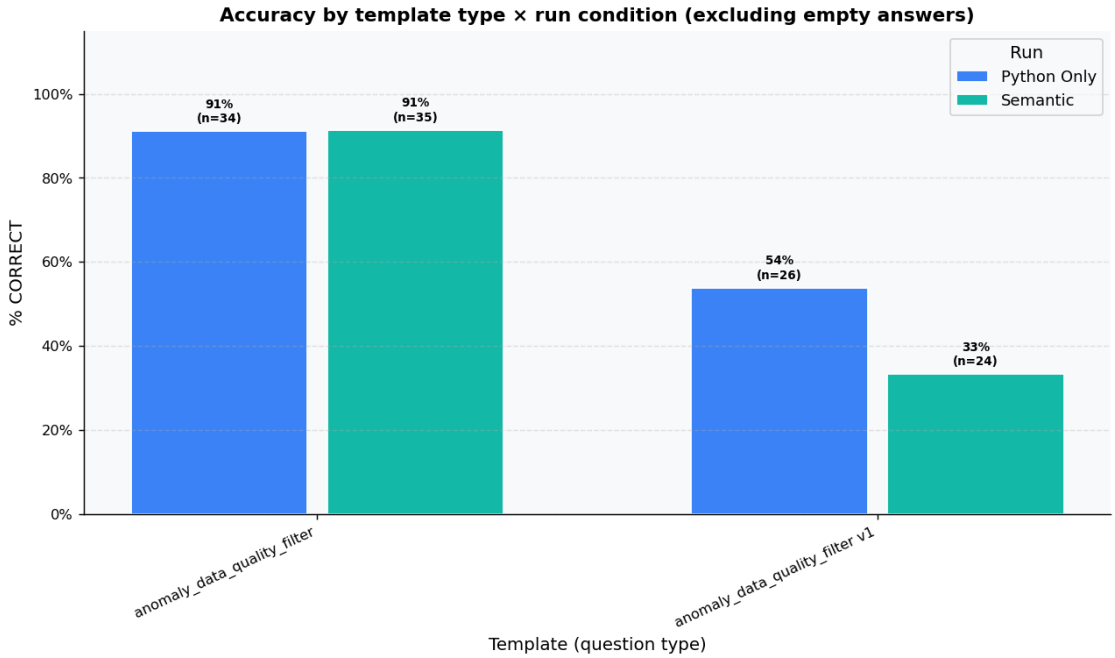



```
Empty answers by model × run:
                                    errors  total rate
model                      run
claude-haiku-4-5-20251001 python_only       3     24  12%
```

```
                          semantic          3     24  12%
claude-opus-4-6           python_only       1     24   4%
                          semantic          5     24  21%
claude-sonnet-4-6         python_only       8     24  33%
                          semantic          5     24  21%

Empty answers by dataset:

                          errors  total  rate rate_fmt
dataset
california_housing_100         3     12  0.25      25%
bike_sharing_100               2     12  0.17      17%
vgsales_100                    1     12  0.08       8%
bike_sharing_500               3     12  0.25      25%
california_housing_500         2     12  0.17      17%
vgsales_500                    2     12  0.17      17%
bike_sharing_1000              4     12  0.33      33%
california_housing_1000        1     12  0.08       8%
vgsales_1000                   0     12  0.00       0%
california_housing_10000       3     12  0.25      25%
vgsales_10000                  3     12  0.25      25%
bike_sharing_10000             1     12  0.08       8%

Datasets with errors: 11 / 12
```

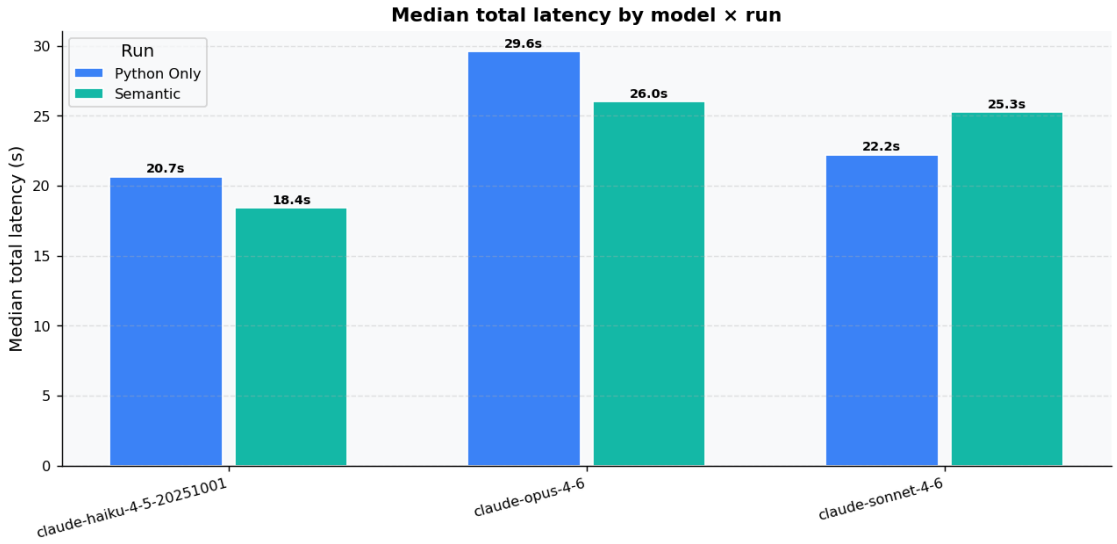


1.8 8 — Template × Run Accuracy

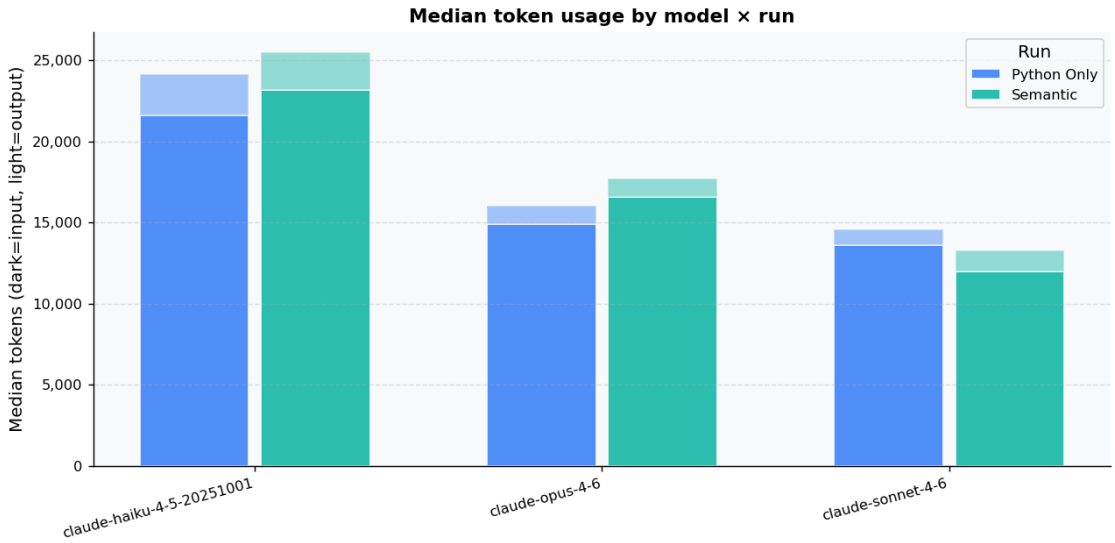


1.9 9 — Latency & Token Metrics

All charts use **median** values (robust to outliers). Error entries are excluded. Metrics: `total_latency_s` (wall-clock), `ttft_s` (time-to-first-token), `input_tokens`, `output_tokens`.



No TTFT data available.



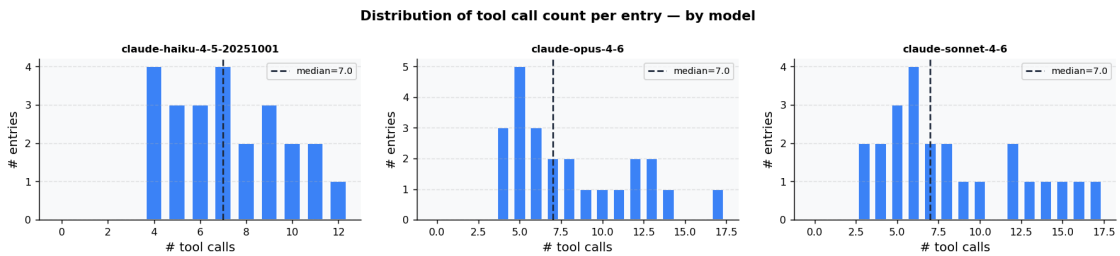
Median metrics per run × model (errors excluded):

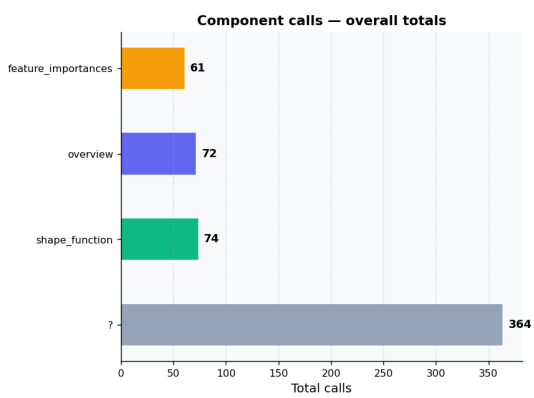
run	model	n	total_latency_s	ttft_s	input_tokens	output_tokens	total_tokens
Python Only	claude-haiku-4-5-20251001	21	20.66	None	21606.00	2564.00	24197.00
	claude-opus-4-6	23	29.59	None	14913.00	1133.00	15860.00
	claude-sonnet-4-6	16	22.23	None	13608.00	974.50	14743.50
Semantic	claude-haiku-4-5-20251001	21	18.42	None	23188.00	2290.00	25369.00
	claude-opus-4-6	19	26.01	None	16597.00	1103.00	17688.00
	claude-sonnet-4-6	19	25.27	None	11974.00	1318.00	13222.00

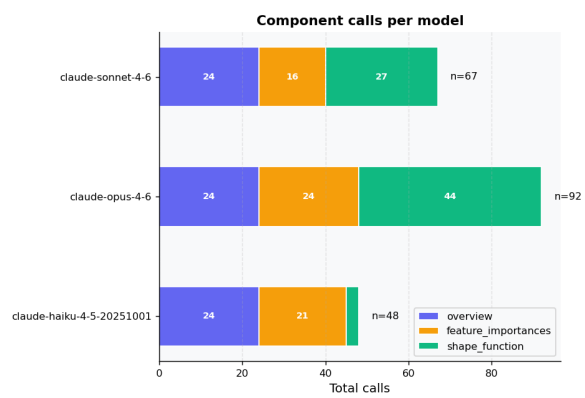
1.10 10 — Semantic Tool Usage Deep Dive

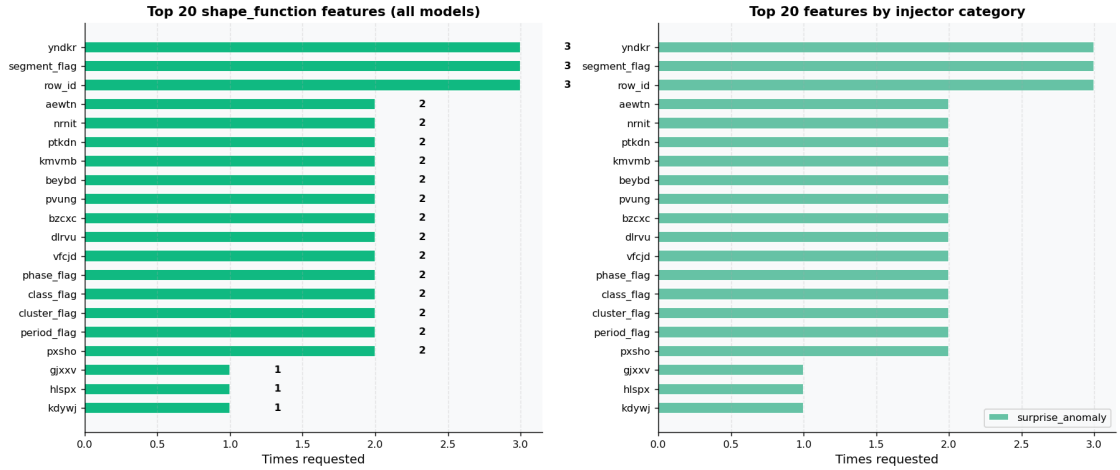
```
Entries in semantic run       : 72
Entries with 0 tool calls     : 0  (0.0%)
Total tool calls made         : 571
Median calls per entry        : 7.0
Max calls per entry           : 17
```

```
Calls per entry by model:
                          median  mean  max  zero_calls
model
claude-haiku-4-5-20251001    7.00  7.25   12           0
claude-opus-4-6              7.00  8.17   17           0
claude-sonnet-4-6            7.00  8.38   17           0

Component breakdown (all entries):
component
?                      364
shape_function          74
overview                72
feature_importances     61
```

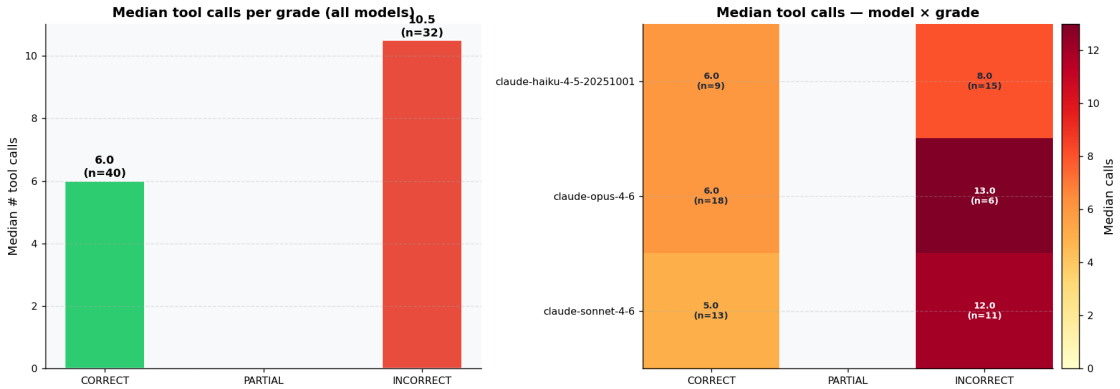









```
posx and posy should be finite values
posx and posy should be finite values
```

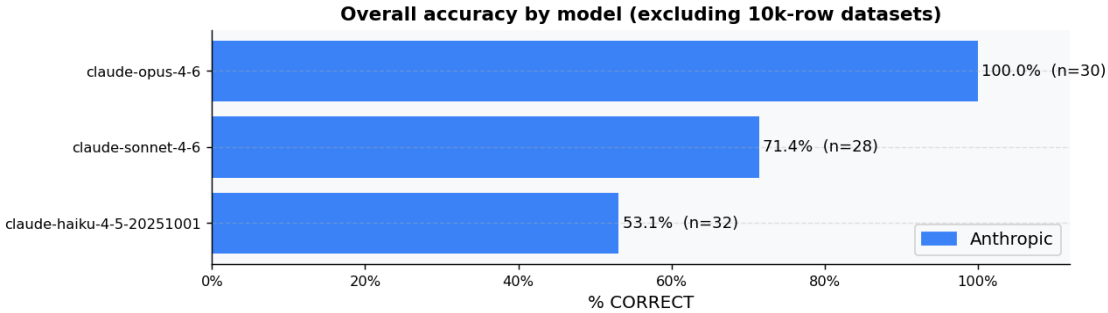


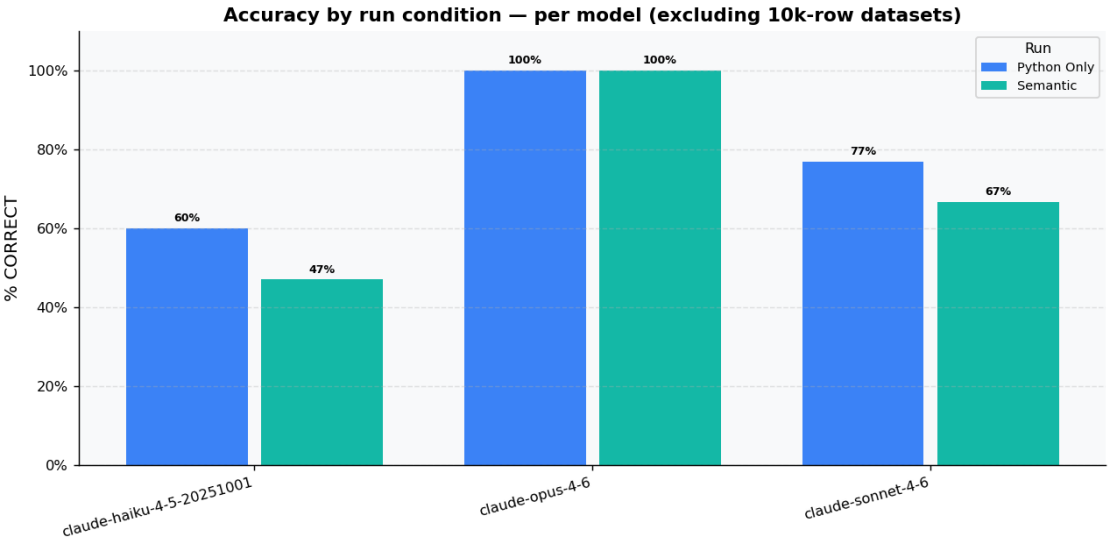

```
Entries where no tools were called: 0

No zero-call entries found.
```

1.11 11 — Performance Excluding 10k-Row Datasets

Since 10k-row datasets cause empty answers for csv_context (29% error rate), this section filters them out entirely so we can compare modes on an equal footing.

```
Records after excluding 10k-row datasets: 90 / 119
Datasets kept: ['bike_sharing_100', 'bike_sharing_1000', 'bike_sharing_500',
'california_housing_100', 'california_housing_1000', 'california_housing_500',
'vgsales_100', 'vgsales_1000', 'vgsales_500']
```


Overall accuracy by model (excluding 10k-row datasets)
claude-opus-4-6
claude-sonnet-4-6
claude-haiku-4-5-20251001
100.0% (n=30)
71.4% (n=28)
53.1% (n=32)
Anthropic
0%
20%
40%
60%
80%
100%
% CORRECT


Accuracy by run condition — per model (excluding 10k-row datasets)
Run
Python Only
Semantic
% CORRECT
100%
80%
60%
40%
20%
0%
60%
47%
100%
100%
77%
67%
claude-haiku-4-5-20251001
claude-opus-4-6
claude-sonnet-4-6